Advanced Functions & Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Solve $\frac{3}{x-1} + 2 = 5$.

$\frac{3}{x-1} = 3 \Rightarrow x - 1 = 1 \Rightarrow x = 2$ (and $x \neq 1$, so this is valid).

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the slant (oblique) asymptote of the rational function: $g(x) = \frac{2x^3 - 5x^2 + 4x + 1}{x^2 - x + 1}$

Step 1: Perform polynomial long division: Dividing $2x^3 - 5x^2 + 4x + 1$ by $x^2 - x + 1$: 1. First term: $2x^3/x^2 = 2x$. Multiply: $2x(x^2 - x + 1) = 2x^3 - 2x^2 + 2x$. Subtract: $(-5x^2 + 4x) - (-2x^2 + 2x) = -3x^2 + 2x + 1$. 2. Second term: $-3x^2/x^2 = -3$. Multiply: $-3(x^2 - x + 1) = -3x^2 + 3x - 3$. Subtract: $(2x + 1) - (3x - 3) = -x + 4$ (remainder). Step 2: Equation of slant asymptote: The quotient is the linear equation: $\mathbf{y = 2x - 3}$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the vertical and horizontal asymptotes of the rational function: $f(x) = \frac{4x^2 - 1}{2x^2 + 5x - 3}$

Step 1: Factor numerator and denominator: $f(x) = \frac{(2x-1)(2x+1)}{(2x-1)(x+3)} = \frac{2x+1}{x+3} \quad \left(x \neq \frac{1}{2}\right)$ Step 2: Vertical asymptote: The non-canceling factor in denominator is $x + 3 = 0 \Rightarrow \mathbf{x = -3}$. (The point at $x = 1/2$ is a removable hole, not a vertical asymptote). Step 3: Horizontal asymptote: Equal degrees (degree 2): ratio of leading coefficients is $y = \frac{4}{2} = \mathbf{2}$.

4 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of the function $f(x) = (x - 1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?

1. Identify the vertex and direction of opening: The function $f(x) = (x - 1)^2 - 4$ is written in vertex form, $f(x) = a(x - h)^2 + k$, where $a = 1$, $h = 1$, and $k = -4$. The vertex of the parabola is at $(1, -4)$. 2. Determine the minimum output value: Since the leading coefficient $a = 1 > 0$, the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by $f(x)$ is -4. 3. Determine the range: The range represents all possible output values (y-values) of the function. Because the function takes on its minimum value of -4 at $x = 1$ and increases without bound as x moves away from 1, the range is all real numbers greater than or equal to -4, which is written in interval notation as $[-4, \infty)$. Common Misconceptions:

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5 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the rational equation: $\frac{2}{x-3} + \frac{1}{x} = \frac{x-1}{x-3}$

Step 1: Domain restrictions: $xeq0$ and $xeq3$. Step 2: Multiply by $x(x-3)$: $2x + 1(x-3) = x(x-1)$ $3x - 3 = x^2 - x \Rightarrow x^2 - 4x + 3 = 0$. Step 3: Factor: $(x-3)(x-1) = 0 \Rightarrow x = 3$ or $x = 1$. Step 4: Check domain: $x = 3$ makes denominators zero (extraneous, rejected). Therefore, the only valid solution is **x = 1**.

6 [Foundational] • Target: Function composition

Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. Find $f(g(2))$ and $g(f(2))$.

$g(2) = 5$, so $f(g(2)) = 3(5) - 2 = 13$. $f(2) = 4$, so $g(f(2)) = 4^2 + 1 = 17$.

7 [Foundational] • Target: Function composition

Using the same f and g, find the formula for $(f \circ g)(x)$.

$(f \circ g)(x) = f(g(x)) = 3(x^2 + 1) - 2 = 3x^2 + 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inverse functions

Find $f^{-1}(x)$ for $f(x) = 3x - 2$.

Let $y = 3x - 2$. Solve for x: $x = \frac{y+2}{3}$. So $f^{-1}(x) = \frac{x+2}{3}$.

9 [Intermediate] • Target: Inverse functions

Verify your answer above by computing $f(f^{-1}(5))$.

$f^{-1}(5) = \frac{5+2}{3} = \frac{7}{3}$, and $f\left(\frac{7}{3}\right) = 3\left(\frac{7}{3}\right) - 2 = 7 - 2 = 5$. It matches, confirming the inverse.

10 [Intermediate] • Target: Rational equations

Solve $\frac{2x-1}{x+3} = \frac{1}{2}$.

Cross-multiply: $2(2x-1) = x + 3 \Rightarrow 4x - 2 = x + 3 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3}$ (and $x \neq -3$, so this is valid).

11 [Intermediate] • Target: Rational equations

Solve $\frac{3}{x-1} + 2 = 5$.

$\frac{3}{x-1} = 3 \Rightarrow x - 1 = 1 \Rightarrow x = 2$ (and $x \neq 1$, so this is valid).

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12 [Intermediate] • Target: Factoring higher-degree polynomials

Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.

By the Factor Theorem, $(x - 1)$ is a factor. Dividing gives $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2)$. Roots: $x = 1,\ 3,\ -2$.

13 [Intermediate] • Target: Factoring higher-degree polynomials

Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.

$(-2)^3 - 2(-2)^2 - 5(-2) + 6 = -8 - 8 + 10 + 6 = 0$. Yes, $x = -2$ is a root.

14 [Intermediate] • Target: Exponential and logarithmic equations

Solve $3^{x-1} = 81$.

$81 = 3^4$, so $x - 1 = 4 \Rightarrow x = 5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Exponential and logarithmic equations • Cross-Lesson: 3.01 Sequences & Series

Solve $\log_3(x + 2) + \log_3(x - 2) = 2$, checking for extraneous roots.

Combine: $\log_3[(x + 2)(x - 2)] = 2 \Rightarrow x^2 - 4 = 3^2 = 9 \Rightarrow x^2 = 13 \Rightarrow x = \pm\sqrt{13}$. The domain requires $x - 2 > 0$, i.e. $x > 2$; since $\sqrt{13} \approx 3.61 > 2$ but $-\sqrt{13} < 2$, only $x = \sqrt{13}$ is valid.

16 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve $\frac{2x-1}{x+3} = \frac{1}{2}$.

Cross-multiply: $2(2x - 1) = x + 3 \Rightarrow 4x - 2 = x + 3 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3}$ (and $x \neq -3$, so this is valid).

17 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x+3}{x-5}$ where $x \neq 5$.

Step 1: Swap x and y: $x = \frac{2y+3}{y-5}$ Step 2: Clear fraction and solve for y:
$x(y - 5) = 2y + 3 \Rightarrow xy - 5x = 2y + 3\ xy - 2y = 5x + 3 \Rightarrow y(x - 2) = 5x + 3\ y = \mathbf{\frac{5x+3}{x-2}}$ $(x \neq 2)$.

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18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Find all real solutions to the radical equation: $\sqrt{2x+15} - x = 0$

Step 1: Isolate radical and square both sides: $\sqrt{2x+15} = x \Rightarrow 2x+15 = x^2 \Rightarrow x^2 - 2x - 15 = 0$ Step 2: Factor: $(x-5)(x+3) = 0 \Rightarrow x = 5 \quad \text{or} \quad x = -3.$ Step 3: Check in original equation: For $x = 5$: $\sqrt{10+15} - 5 = \sqrt{25} - 5 = 0$ ✓. For $x = -3$: $\sqrt{-6+15} - (-3) = \sqrt{9} + 3 = 3 + 3 = 6eq0$ (extraneous). Thus the unique solution is **$x = 5$**.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

If r_1 and r_2 are the roots of $3x^2 - 8x + 5 = 0$, find the exact value of $\frac{1}{r_1} + \frac{1}{r_2}$ without solving for the roots individually.

Step 1: Combine into common denominator: $\frac{1}{r_1} + \frac{1}{r_2} = \frac{r_1 + r_2}{r_1 r_2}$ Step 2: Apply Vieta's formulas: Sum of roots: $r_1 + r_2 = -\frac{b}{a} = -\frac{-8}{3} = \frac{8}{3}$. Product of roots: $r_1 r_2 = \frac{c}{a} = \frac{5}{3}$. Step 3: Divide: $\frac{r_1 + r_2}{r_1 r_2} = \frac{8/3}{5/3} = \mathbf{\frac{8}{5}}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

Convert the repeating decimal $0.\overline{45}$ to an irreducible common fraction.

Step 1: Express as an infinite geometric series: $0.454545\ldots = \frac{45}{100} + \frac{45}{10000} + \cdots$ where $a_1 = \frac{45}{100}$ and $r = \frac{1}{100}$. Step 2: Sum the series: $S_\infty = \frac{\frac{45}{100}}{1 - \frac{1}{100}} = \frac{\frac{45}{100}}{\frac{99}{100}} = \frac{45}{99} = \mathbf{\frac{5}{11}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $\frac{4x-3}{x-8} = 5$.

a. 38

b. 36

c. 37 ✓

d. -37

Cross-multiply: $4x - 3 = 5(x - 8)$, so $-x = -37$, giving $x = 37$ (and $x \neq 8$, so this is valid).

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22 [Foundational] • Target: Inverse functions

Let $f(x) = x - 9$ and $g(x) = -3x^2 + 7$. Find $f(g(0))$.

a. 0

b. 7

c. -4

d. -2 ✓

$g(0) = -3(0)^2 + 7 = 7$. $f(7) = (7) - 9 = -2$.

23 [Foundational] • Target: Rational equations

Let $f(x) = -2x - 6$ and $g(x) = -x^2 - 1$. Find $f(g(-4))$.

a. 28 ✓

b. -17

c. 26

d. 30

$g(-4) = -(-4)^2 - 1 = -17$. $f(-17) = -2(-17) - 6 = 28$.

24 [Foundational] • Target: Factoring higher-degree polynomials

Let $f(x) = x + 7$ and $g(x) = x^2 - 8$. Find $f(g(-5))$.

a. 26

b. 24 ✓

c. 17

d. 22

$g(-5) = (-5)^2 - 8 = 17$. $f(17) = (17) + 7 = 24$.

25 [Foundational] • Target: Exponential and logarithmic equations

Let $f(x) = 4x + 4$ and $g(x) = -3x^2 - 7$. Find $f(g(-4))$.

a. -218

b. -55

c. -214

d. -216 ✓

$g(-4) = -3(-4)^2 - 7 = -55$. $f(-55) = 4(-55) + 4 = -216$.

26 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $f(x) = -3x + 3$ and $g(x) = -3x^2 + 3$. Find $f(g(1))$.

a. 1

b. 0

c. 3 ✓

d. 5

$g(1) = -3(1)^2 + 3 = 0$. $f(0) = -3(0) + 3 = 3$.

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27 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $f(x) = -4x - 9$ and $g(x) = -4x^2 - 3$. Find $f(g(-3))$.

a. -39

b. 145

c. 147 ✓

d. 149

$g(-3) = -4(-3)^2 - 3 = -39$. $f(-39) = -4(-39) - 9 = 147$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Let $f(x) = 2x + 3$ and $g(x) = 4x^2 - 8$. Find $f(g(5))$.

a. 187 ✓

b. 185

c. 189

d. 92

$g(5) = 4(5)^2 - 8 = 92$. $f(92) = 2(92) + 3 = 187$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve $\frac{2x-4}{x+4} = -2$.

a. -1 ✓

b. 0

c. 1

d. -2

Cross-multiply: $2x - 4 = -2(x + 4)$, so $4x = -4$, giving $x = -1$ (and $x \neq -4$, so this is valid).

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $\frac{3x+4}{x-9} = -2$.

a. $-\frac{14}{5}$

b. $\frac{14}{5}$ ✓

c. $\frac{9}{5}$

d. $\frac{19}{5}$

Cross-multiply: $3x + 4 = -2(x - 9)$, so $5x = 14$, giving $x = \frac{14}{5}$ (and $x \neq 9$, so this is valid).

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $\frac{x+9}{x-2} = -4$.

a. $-\frac{1}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. $-\frac{6}{5}$

Cross-multiply: $x + 9 = -4(x - 2)$, so $5x = -1$, giving $x = -\frac{1}{5}$ (and $x \neq 2$, so this is valid).

32 [Intermediate] • Target: Function composition

Solve $\frac{-4x+5}{x-2} = -3$.

a. 0

b. 1

c. -2

d. -1 ✓

Cross-multiply: $-4x + 5 = -3(x - 2)$, so $-x = 1$, giving $x = -1$ (and $x \neq 2$, so this is valid).

33 [Intermediate] • Target: Inverse functions

Solve $\frac{-x+2}{x-3} = 3$.

a. $\frac{7}{4}$

b. $\frac{11}{4}$ ✓

c. $-\frac{11}{4}$

d. $\frac{15}{4}$

Cross-multiply: $-x + 2 = 3(x - 3)$, so $-4x = -11$, giving $x = \frac{11}{4}$ (and $x \neq 3$, so this is valid).

34 [Intermediate] • Target: Rational equations

Solve $\frac{-x+9}{x-1} = 2$.

a. $\frac{11}{3}$ ✓

b. $\frac{14}{3}$

c. $\frac{8}{3}$

d. $-\frac{11}{3}$

Cross-multiply: $-x + 9 = 2(x - 1)$, so $-3x = -11$, giving $x = \frac{11}{3}$ (and $x \neq 1$, so this is valid).

35 [Intermediate] • Target: Factoring higher-degree polynomials

Find $f^{-1}(x)$ for $f(x) = 2x - 5$.

a. $\frac{x+5}{2}$ ✓

b. $\frac{x+5}{-2}$

c. $2x - 5$

d. $\frac{x-5}{2}$

Let $y = 2x - 5$. Solve for x: $x = \frac{y+5}{2}$. So $f^{-1}(x) = \frac{x+5}{2}$.

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36 [Intermediate] • Target: Exponential and logarithmic equations

Find $f^{-1}(x)$ for $f(x) = 4x + 7$.

a. $\frac{x-7}{-4}$

b. $4x + 7$

c. $\frac{x-7}{4}$ ✓

d. $\frac{x+7}{4}$

Let $y = 4x + 7$. Solve for x: $x = \frac{y-7}{4}$. So $f^{-1}(x) = \frac{x-7}{4}$.

37 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given that $x = 4$ is a root of $x^3 - 13x - 12 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -1)$ ✓

b. $(3, 1)$

c. $(-2, -1)$

d. $(-3, -2)$

Dividing out $(x - 4)$ gives $x^2 + 4x + 3 = 0$, which factors as $(x - (-1))(x - (-3)) = 0$, so $x = -1$ or $x = -3$. Answer: $(-3, -1)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Complex Numbers

Given that $x = 3$ is a root of $x^3 - 10x^2 + 27x - 18 = 0$, use the Factor Theorem to find the other two roots.

a. $(2, 6)$

b. $(1, 5)$

c. $(-1, -6)$

d. $(1, 6)$ ✓

Dividing out $(x - 3)$ gives $x^2 - 7x + 6 = 0$, which factors as $(x - 1)(x - 6) = 0$, so $x = 1$ or $x = 6$. Answer: $(1, 6)$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

Given that $x = 2$ is a root of $x^3 + 6x^2 - 9x - 14 = 0$, use the Factor Theorem to find the other two roots.

a. $(-6, -1)$

b. $(-7, -2)$

c. $(7, 1)$

d. $(-7, -1)$ ✓

Dividing out $(x - 2)$ gives $x^2 + 8x + 7 = 0$, which factors as $(x - (-7))(x - (-1)) = 0$, so $x = -7$ or $x = -1$. Answer: $(-7, -1)$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Sequences & Series

Solve $5^{2x+7} = 5$ for x.

a. -4

b. -2

c. -1

d. -3 ✓

Since $5 = 5^1$: $2x + 7 = 1$, so $2x = -6$, giving $x = -3$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve $2^x = 8$ for x.

a. 5

b. 2

c. 3 ✓

d. 4

Since $8 = 2^3$: $x = 3$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Limits

Solve $\frac{3x-6}{x+1} = -4$.

a. $\frac{9}{7}$

b. $\frac{2}{7}$ ✓

c. $-\frac{5}{7}$

d. $-\frac{2}{7}$

Cross-multiply: $3x - 6 = -4(x + 1)$, so $7x = 2$, giving $x = \frac{2}{7}$ (and $x \neq -1$, so this is valid).

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Complex Numbers

Solve $\frac{4x+6}{x-7} = -4$.

a. $\frac{11}{4}$ ✓

b. $-\frac{11}{4}$

c. $\frac{15}{4}$

d. $\frac{7}{4}$

Cross-multiply: $4x + 6 = -4(x - 7)$, so $8x = 22$, giving $x = \frac{11}{4}$ (and $x \neq 7$, so this is valid).

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Solve the rational inequality: $\frac{x-3}{x+2} \leq 0$.

a. $(-2, 3]$ ✓

b. $[-2, 3]$

c. $(-\infty, -2) \cup [3, \infty)$

d. $x \leq 3$

Critical values at $x = -2$ and $x = 3$. Testing intervals shows the expression is negative on $(-2, 3)$. Note $x = -2$ is excluded because it makes denominator zero.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

What is the form of the partial fraction decomposition for $\frac{3x+1}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$ ✓

b. $\frac{A}{x-1} + \frac{B}{x^2+4}$

c. $\frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-2}$

d. $\frac{Ax+B}{x-1} + \frac{C}{x^2+4}$

For an irreducible quadratic factor in denominator, the numerator must be a general linear polynomial: $Bx + C$.

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